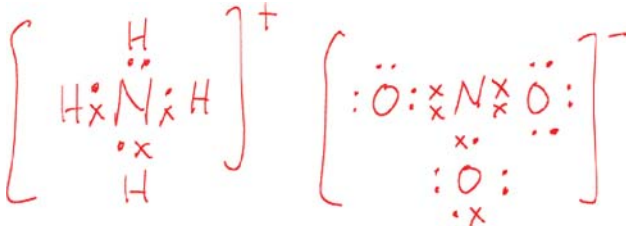
Paper 2 Answers

1 (a) (i)



Shape of ammonium ion: tetrahedral
Shape of nitrate ion: trigonal planar

(ii) ΔH_{sol} = -(-Lattice Energies) + ($-\Delta H_{hyd}$) = |L.E.| - | ΔH_{hyd} |

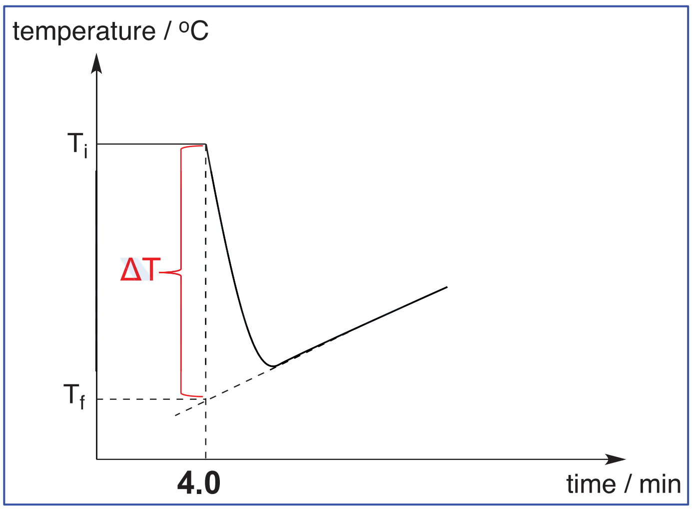
$|L.E.| \propto \frac{|q^+q^-|}{r^+ + r^-}$

$|\Delta H_{hyd}| \propto \frac{|q|}{r}$

Anionic radius of nitrate ion is larger than chloride, therefore the decrease in | ΔH_{hyd} | of nitrate ion is larger than the decrease in |L.E.|. ΔH_{sol} is expected to be more endothermic.

(b) (i) Assuming a temperature change of 5 °C and no heat loss to surroundings,
n(salt) x 15 000 = 100 x 4.3 x 5
n(salt) = 0.1433 mol
minimum mass = 0.1433 x 53.5 = 7.67 g

(ii)



(iii) $\Delta G^o = \Delta H^o - T\Delta S^o$
ΔG^o is negative since reaction is spontaneous and ΔH^o is positive since reaction is endothermic. Therefore sign of ΔS^o is positive as there are more ways to arrange the particles when the solid dissolves in water.

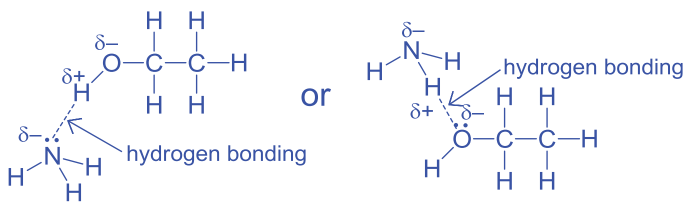
(c) **(i)** Since Zn is oxidised, $Zn(s) \rightarrow Zn^{2+}(aq) + 2\,e^-$

Overall equation:
$Zn(s) + 2\,MnO_2(s) + 2\,NH_4Cl(aq) \rightarrow Mn_2O_3(s) + ZnCl_2(aq) + 2NH_3(aq) + H_2O(l)$

(ii) $E^\circ_{cell} = (+0.5) - (-0.76) = +1.26V$

(iii) $\Delta G^o = -nFE^o = -2 \times 96500 \times 1.26 = -243\,000\ J\ mol^{-1}$
$= -243\ kJ\ mol^{-1}$
The sign of ΔG^o is negative and hence the reaction is spontaneous.

2 **(a)**



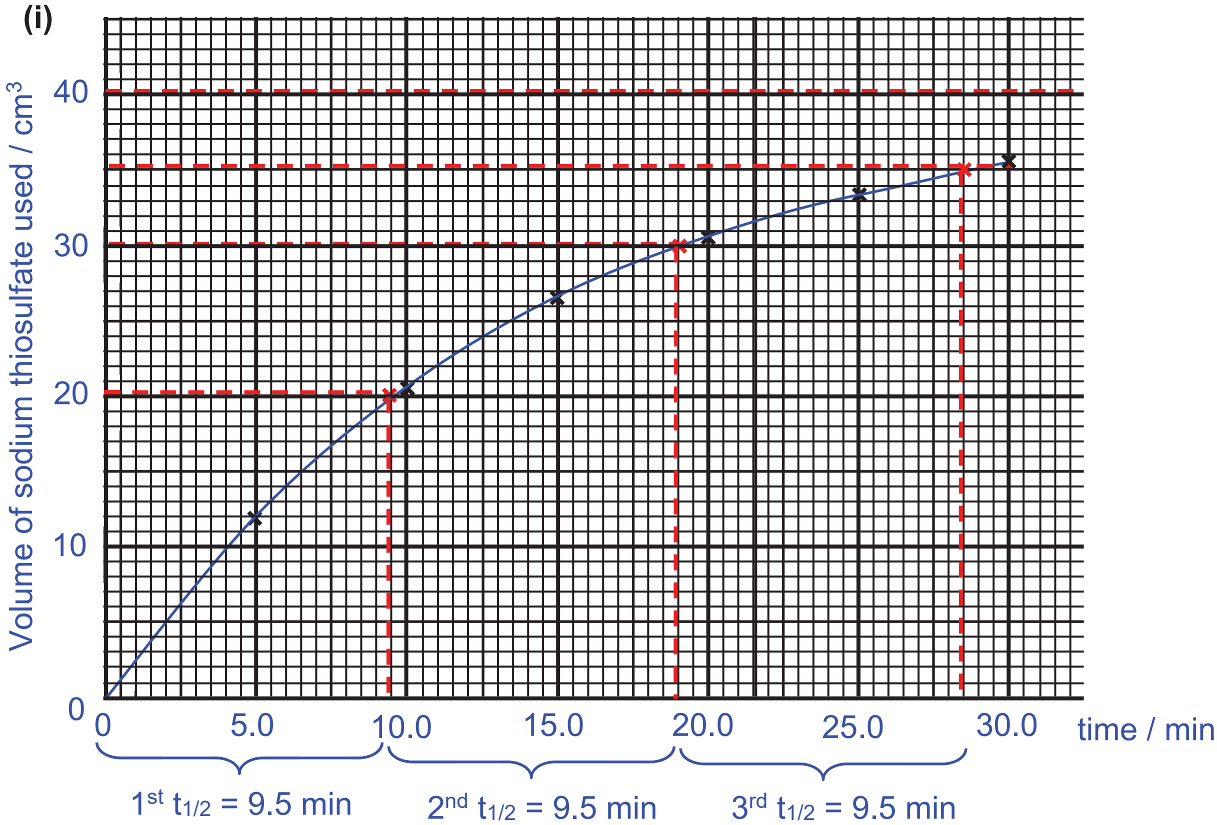
(b) **(i)** Quenching is required to stop or slow down the reaction so as to achieve a more accurate titre value at that time or to find the concentration at that instance.
Quenching agent: large volume of cold water / add large volume of acid to remove the NH_3 (in this question).

(ii) $Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$

(iii) $I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$

(iv) Starch indicator is added when the solution turns pale yellow.
The end-point can be recognised when one drop of sodium thiosulfate added cause the dark blue solution to permanently turn colourless.

(c) (i)



(ii) [2-iodobutane] = 0.20 / 2 = 0.10 mol dm^{-3}

n(2-iodobutane) : n(I^-) : n($S_2O_3^{2-}$) = 1 : 1 : 1

n(2-iodobutane) = n($S_2O_3^{2-}$) = $\frac{10}{1000}$ × 0.10 = 0.001000 mol

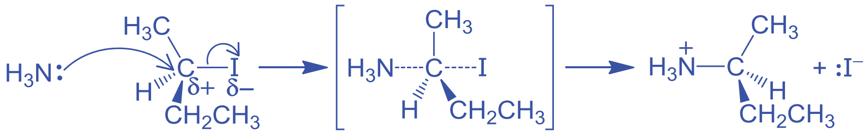
V($S_2O_3^{2-}$) required when all 2-iodobutane reacted = $\frac{0.001000}{0.0250}$ = 0.04000 dm^3 = 40.00 cm^3

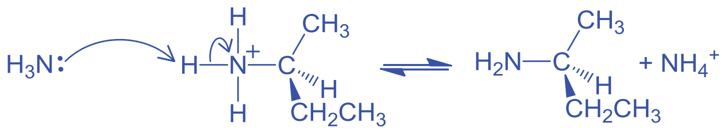
When volume of sodium thiosulfate increases from 0 to 20 cm^3,
1st $t_{1/2}$ = 9.5 min .
When volume of sodium thiosulfate increases from 20 to 30 cm^3
2nd $t_{1/2}$ = 9.5 min.
Since the 1st $t_{1/2}$ is approximately equal to 2nd $t_{1/2}$, it is 1st order with respect to 2-iodobutane.

(iii) Since the rate for 2.00 mol dm^{-3} of ethanolic ammonia reaction is half the rate for 4.00 mol dm^{-3} of ethanolic ammonia reaction, it is 1st order with respect to ethanolic ammonia.

(iv) rate = k [2-iodobutane][ethanolic ammonia]

(v) Nucleophilic Substitution (S_N2)





(vi) After mixing equal volume of 0.20 mol dm^{-3} 2-iodobutane and 4.00 mol dm^{-3} ethanolic ammonia, [2-iodobutane] = 0.10 mol dm^{-3} and [ethanolic ammonia] = 2.00 mol dm^{-3}.

Since [ethanolic ammonia] is in large excess,
rate = k' [2-iodobutane], where k' = k [ethanolic ammonia]

$t_{1/2} = \frac{\ln 2}{k'} = \frac{\ln 2}{k\text{ [ethanolic ammonia]}}$

$9.5 = \frac{\ln 2}{k\,(2.00)}$

k = 0.0365 [1] mol^{-1} dm^{3} min^{-1}

(vii) I. 2-iodobutane is replaced with 2-chlorobutane.

BE(C−C*l*) = 340 kJ mol^{-1} BE(C−I) = 240 kJ mol^{-1}

As <u>more energy is required to overcome the stronger C−C*l* bond</u>, the rate of reaction <u>decreases</u>.

II. ethanolic ammonia is replaced with ethanolic ethylamine.

<u>Electron donating ethyl group increases the electron density</u> around the nitrogen atom of ethylamine, making the <u>lone pair of electrons more available</u> (or make the nitrogen <u>more nucleophilic</u>) to attack the electrophilic carbon. Hence, rate of reaction <u>increases</u>.

3 **(a)** From Cr to Co,

- number of protons increases, nuclear charge increases
- additional electron is added to the penultimate 3d subshell.
- Hence, screening/shielding effect also increases as presence of the 3d orbital shields the 4s electrons from the nuclear attraction.
- The effective nuclear charge experience by the outer 4s electrons increases only very gradually.
- Energy required to remove the 4s electron is relatively invariant.

(b) **(i)** Ksp = $[Cr^{3+}][OH^-]^3$
$6.3 \times 10^{-31} = (0.010/2)[OH^-]^3$
$[OH^-] = 5.013 \times 10^{-10}$ mol dm^{-3}

(ii) Ksp = $[Fe^{3+}][OH^-]^3$
$4 \times 10^{-38} = [Fe^{3+}](5.013 \times 10^{-10})^3$
$[Fe^{3+}] = 3.174 \times 10^{-10}$ mol dm^{-3} << 0.005 mol dm^{-3} hence effective

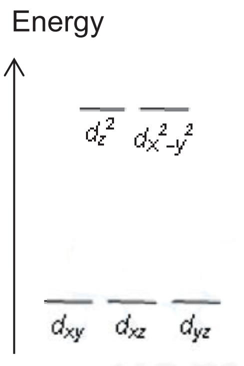
(iii) To the solution of Cr^{3+} and Fe^{3+} ions, add sodium hydroxide until excess. $Cr(OH)_3$ ppt forms and is soluble in excess.

$Cr^{3+} + OH^- \rightarrow Cr(OH)_3$
$Cr(OH)_3 + 3OH^- \rightarrow [Cr(OH)_6]^{3-}$

$Fe(OH)_3$ ppt is insoluble in excess sodium hydroxide.
$Fe^{3+} + OH^- \rightarrow Fe(OH)_3$

Filter the mixture and $Fe(OH)_3$ is the residue and $[Cr(OH)_6]^{3-}$ is the filtrate.

(c) **(i)** The electrons in orbitals that lie along the same axes as the ligands experiences greater repulsion, hence the energy is raised.



(ii) In the presence of ligands, the 3d orbitals split into 2 groups with an energy gap. When visible light passes through the iron complex, the violet wavelength of light corresponding to the energy gap is absorbed by the 3d electron in the lower energy level. This electron is promoted to a vacant 3d orbital at the higher energy level. The complementary colour, corresponding to unabsorbed wavelengths is observed.

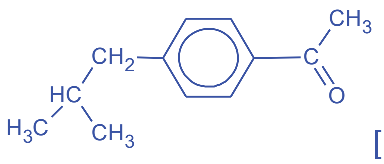
4 **(a)** At low pressure, when volume increases, pressure of chloromethane falls more than that of ideal gas as <u>permanent dipole-permanent dipole interaction</u> between CH_3Cl molecules hold the particles closer together, hence they strike the walls of the container with less force, resulting in lower pressure.

OR

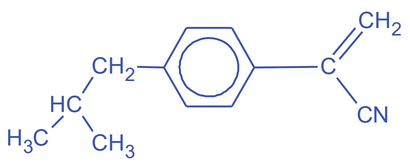
At low pressure, volume of chloromethane gas is lower/decreases more than ideal gas for a given pressure as <u>permanent dipole-permanent dipole interaction</u> between molecules is significant and the molecules are attracted closer to each other.

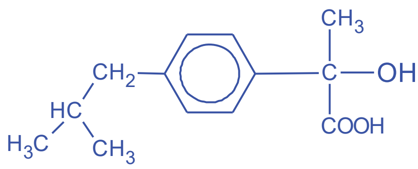
(b) **(i)** Electrophilic Substitution.

(ii)

A: 

[1]

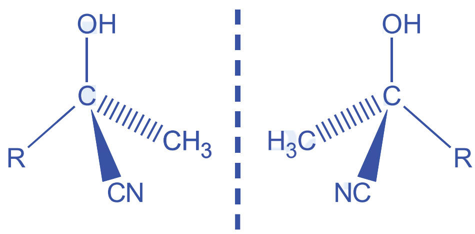
B: 

or 

(iii)

	Reagents and conditions	
Step I	anhydrous $AlCl_3$,$(CH_3)_2CHCH_2Cl$, room temperature	
Step IV	Al_2O_3, heat at 350 °C (OR conc H_2SO , 170 °C less preferred as hydrolysis of nitrile may occur)	Any dilute acid, heat under reflux
Step VI	H_2, Ni catalyst, heat **OR** H_2, Pt / Pd catalyst, room temp.	

(iv)



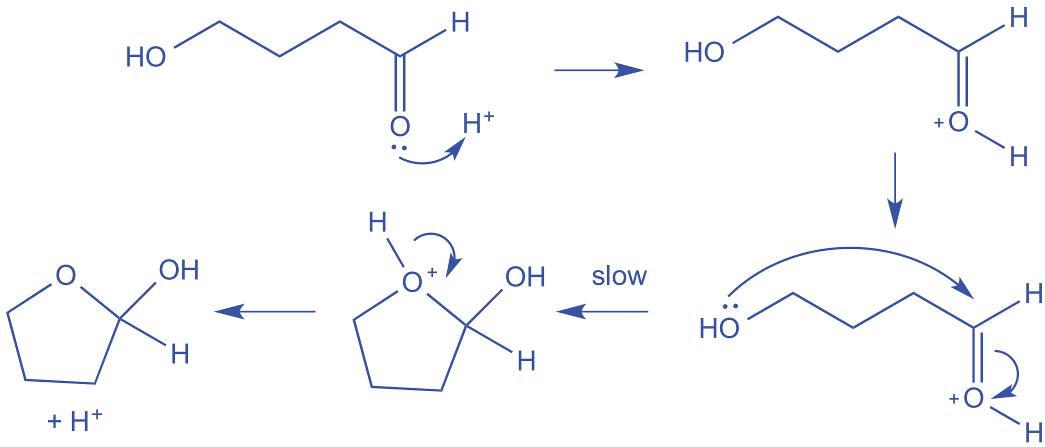
The <u>CN^- nucleophile</u> can approach the trigonal <u>planar carbonyl carbon from the top or bottom of the plane</u> **with equal probability** to <u>give the 2 enantiomers</u> **in equal amounts**.

(v) The <u>sp^3 chiral carbon</u> formed <u>in step III</u> eventually became <u>sp^2 hybridised after elimination in step V (or IV)</u> hence the two enatiomers will form the same alkene (which doesn't exhibit cis-trans isomerism).

(c) The relative strength of an acid depends on the stability of its conjugate base formed. Methoxyphenol has a lower pK_a, thus has a higher K_a value and is a stronger acid than methylphenol.
The <u>-OCH_3 group is slightly more electronegative</u> than CH_3 group [1] thus decreases the electron density of the conjugate base and <u>disperses the negative charge on the phenoxide O atom to the OCH_3 group</u> [1] hence has greater stability than the methylphenoxide.

Alternatively:
The relative strength of an acid depends on the stability of its conjugate base formed. Methoxyphenol has a lower pK_a, thus has a higher K_a value and is a stronger acid than methylphenol.
The p orbital of O in OCH_3 group overlaps with the π orbital of the phenoxide ion, thus there is greater dispersion of the negative charge over a larger volume [1] hence methoxyphenol has greater stability than the methylphenoxide.

5 **(a)** **(i)** Nucleophilic Addition

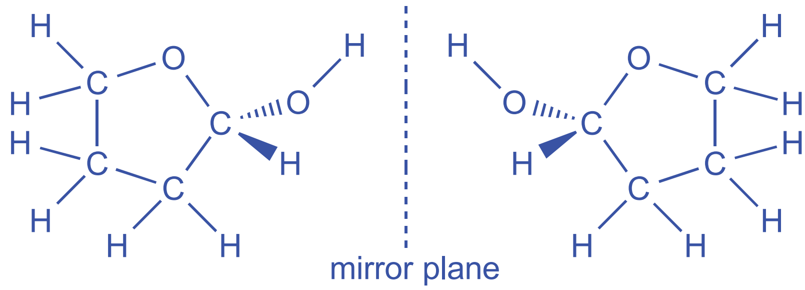


(ii) Add 2,4-DNPH, warm (Accept Fehling's solution OR Tollens' reagent)

When an orange precipitate seen, 4-hydroxyaldehyde is present. The formation of the cyclic hemiacetal is reversible.

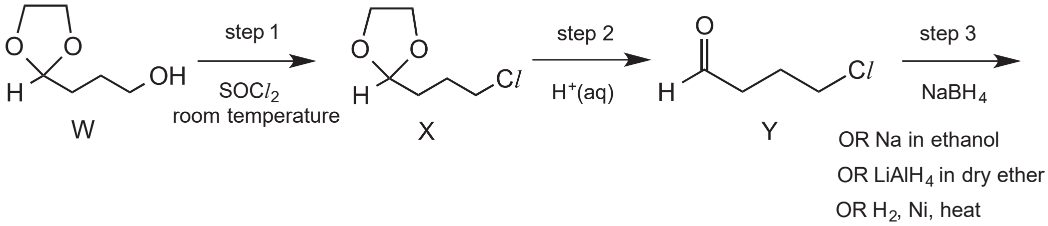
No orange precipitate seen, 4-hydroxyaldehyde is absent. The formation of cyclic hemiacetal is irreversible.

(iii)

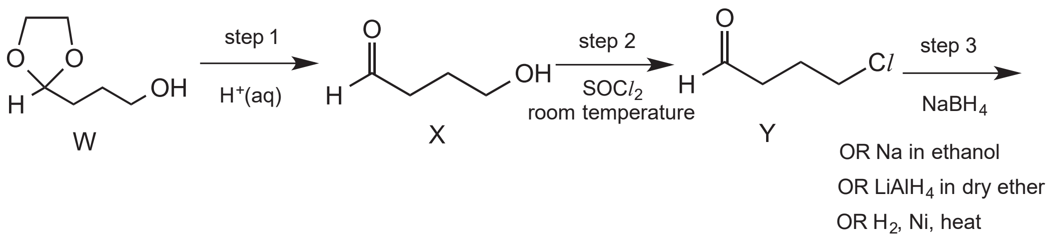


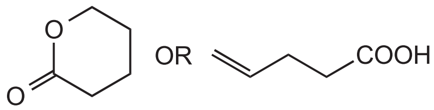
(b) (i) The electron donating $-OCH_3$ in ester disperses the partial positive charge on the carbonyl carbon, making it less electrophilic. Hence less susceptible to attack by hydride.

(ii)
(iii) Possible Synthetic route 1:



Possible Synthetic route 2:





Z

(iv) step 4: nucleophilic substitution
step 5: acid hydrolysis or hydrolysis